Worksheet 1 — Security Mindset & Threat Modeling (3 hrs)

> **Course:** Software Security (1305315) · **Week 1**
> **Aligned to:** OWASP 2025 A06 Insecure Design · CWE-501 (Trust Boundary Violation)
> **Signature game:** "Elevation of Privilege" (Microsoft STRIDE card deck)
>
> **Ethics note:** modeling-only week. The one before/after in Task 8 is the worksheet's own defense-verification step, run against the instructor-provided sample app on `localhost:8081` only, under `ETHICS.md`.

Part 1 — Student Information

Name	Student ID	Date	Group
Thiha Lin	6631503092	2026-08-15	*(team TBD — set before Week 4)*

Part 2 — Lecture Questions

1. CIA triad + one failure each.
Confidentiality, Integrity, Availability — the three properties security protects. *Confidentiality* (only authorized parties read data): `GET /notes` returns every user's notes to any caller, so one user reads another's. *Integrity* (no unauthorized modification): `/upload` with `../pwned.txt` lets an attacker overwrite an arbitrary file on the server. *Availability* (usable when needed): no upload-size or rate limit means one caller can fill the disk and deny everyone else.

2. Trust boundary + why scrutiny.
A trust boundary is a line where the trust level of data changes — typically where data crosses from a less-trusted zone into a more-trusted one (Internet → app). Data crossing it deserves extra scrutiny because on the far side it gets acted on as if trustworthy; every assumption you hold inside (well-formed, authorized, size-bounded) must be *re-established by validation at the crossing*, since the untrusted sender controls the data.

3. Attack surface + two things that grow it.
The attack surface is the sum of all points where an attacker can send data to or interact with the system — every endpoint, parameter, header, and file input. It grows when you (a) add endpoints or parameters that accept user input (e.g. an upload route that trusts a filename), and (b) pull in third-party dependencies — each adds code you didn't write and CVEs you inherit.

4. STRIDE → threat → property violated.
Spoofing → impersonation → violates **Authentication**. Tampering → unauthorized modification → **Integrity**. Repudiation → denying an action with no proof → **Non-repudiation**. Information disclosure → exposing data → **Confidentiality**. Denial of service → resource exhaustion → **Availability**. Elevation of privilege → gaining rights you shouldn't → **Authorization**.

5. Secure by Design (CISA) vs bolting on.
Secure by Design means building security into the architecture and defaults from the start — safe defaults, least privilege, threat-modeling before code — so the product ships secure and the burden isn't pushed onto the customer. Bolting on after release patches symptoms around a design that already assumed trust; designed-in security instead removes whole vulnerability *classes* by construction (parameterized queries make SQLi structurally impossible rather than filtered case-by-case).

Part 3 — Hands-on Lab

Per the worksheet: each task gives the threat/element identified, a screenshot, and a 2–3 sentence mitigation. Screenshots carry the identity stamp (see Evidence & Integrity).

Task 0 — Onboarding. ✅ App runs at `http://localhost:8081` (host 8080 is taken by another local service, so the lab is remapped). `GET /notes` → `[]`; after a seed POST → `[[1,"alice","hello]]`; `POST /upload` → `{"saved":"demo.txt"}`; `GET /files/demo.txt` returns the content. **Screenshot:** terminal (identity stamp) beside the JSON responses. *(attach)*

Task 1 — DFD. *Element:* whole system. **Screenshot/diagram:** `img/dfd-6631503092.png`, embedded at `threat-model-6631503092.md §1`. *Mitigation:* the diagram itself shows the single Internet→app trust boundary carries no check; the structural mitigation is to make that crossing an enforcement point (authenticate, validate, rate-limit) rather than a line data simply passes through. Everything else in this worksheet is downstream of that one gap.

Task 2 — STRIDE grid. *Elements:* `/notes`, `/upload`, `/files/<name>`. **Screenshot:** the completed grid at `threat-model-6631503092.md §3`. *Mitigation:* the grid's dominant column is Spoofing/EoP caused by the total absence of authentication, so introducing sessions and deriving `owner` server-side closes the largest number of cells at once. Repudiation is universal and closes with one control: structured request logging.

Task 3 — Elevation of Privilege game. *Carded threats tied to real elements (score = 7):* Spoofing→ `/notes` client-supplied `owner`; Tampering→ `/upload` `../`; Repudiation→no logs (all elements); Info-disclosure→ `/upload` echoes the save path; Info-disclosure→ `GET /notes` returns everyone's notes; DoS→no size/rate limit; EoP→write into `/app` via traversal. **Screenshot:** deck + recorded threats. *(attach) Mitigation:* the deck surfaced nothing outside the grid's classes, which raises confidence the grid is complete at element level — the gap it could not surface is chaining, which is why Task 3b exists.

Task 3b — Systems-level pass. *Element:* whole diagram. **Screenshot:** trust-boundary simulation + written pass at `threat-model-6631503092.md §4`. *Mitigation:* per-element fixes do not address the two system properties (running as root, no authn boundary), so the mitigation is architectural — drop privileges to a non-root user and put an authentication boundary in front of every route, so no single future element bug is total.

Task 4 — Abuse cases & personas. *Elements:* `/upload`, `/notes`, `notes.db`. **Screenshot:** `threat-model-6631503092.md §5`. *Mitigation:* both personas succeed through the same root cause — the app never establishes *who* is calling — so per-request authentication plus server-derived ownership defeats AC-1 through AC-3; AC-4 (disk-fill) additionally needs upload-size and rate limits.

Task 5 — Path-traversal deep-dive. *Element:* `/upload` → `uploads/` → `/files/<name>`. **Screenshot:** trace + secure design at `threat-model-6631503092.md §6`. *Mitigation:* `secure_filename()` strips separators and `..` so the name cannot climb out of `uploads/`; add an extension allow-list, store uploads outside the app directory, and run non-root so that even a future write-primitive cannot reach code. Implemented in Task 8.

Task 6 — NoteVault (term-project target). *Elements:* `/login`, `/register`, `/api/notes/<id>`. **Screenshot:** NoteVault running on `http://localhost:8082` + top-3 at `threat-model-6631503092.md §7`. *Mitigation:* parameterise the `/login` query (kills the SQLi bypass), pin the JWT algorithm to HS256 and drop `"none"` from the decode allow-list (kills token forgery), and bind `role` server-side instead of accepting it from the client (kills the mass-assignment path to admin).

Task 7 — Security requirements. *Elements:* all. **Screenshot:** `threat-model-6631503092.md §8`. *Mitigation:* the three requirements are written as testable acceptance criteria, so each becomes a check that can fail a build rather than a note in a document — which is what makes a mitigation durable rather than a one-time patch.

Task 8 — Defend / fix it. ✅ **Implemented.** Top-5 ranking, the before/after evidence and the class-vs-instance discussion are at `threat-model-6631503092.md §9`. *Mitigation:* `secure_filename()` + allow-list on `/upload`.
- **Commit:** `c254723` — https://github.com/Thiha-Lynn/software-security/commit/c254723
- **Pull request:** https://github.com/Thiha-Lynn/software-security/pull/1

Part 4 — Reflection

1. Top finding → CWE + OWASP A06.
The `/upload` arbitrary file write is **CWE-22** (path traversal) / **CWE-501** (trust-boundary violation), mapping to **A06 Insecure Design** because the flaw is architectural — the design *trusted a client-supplied filename as a safe path component*, so untrusted data crossed the Internet→app boundary and became a filesystem path with no re-validation.

2. Real-world breach from a design flaw (not a missing patch).
Capital One, 2019 — an SSRF let an attacker reach the cloud metadata service and assume an **over-permissioned instance role that could read every S3 bucket**, exposing ~100M records. The root cause was design, not an unpatched CVE: excessive privilege plus a reachable metadata endpoint. The design control that prevents it is **least privilege** (scope the role to the one bucket it needs) — the same principle as my system claim that the sample app must stop running as root.

3. Best risk reduction per unit of effort.
`secure_filename()` on `/upload` — ~4 lines that close a *Critical* class (arbitrary file write, which chains to RCE). Authentication removes more total risk but is a much larger change; the traversal fix is the highest reduction-per-line, so it's the one I implemented first.

Evidence & Integrity

- **Identity stamp** (run this and keep it in the same image as every screenshot):
  `printf '%s | %s | ' "$(whoami)" '6631503092'; date '+%F %T %Z'`
- **Personalized flag:** *none found in the Week 1 lab materials; to be confirmed on `learn.zcr.ai` at submission. Task 8's before/after evidence is this lab's proof artifact.*
- **Explain in your own words:**
  1. *What I did & why it worked:* I sent `POST /upload` with `filename=../pwned_before.txt`. The handler did `f.save(os.path.join("uploads", f.filename))`, and `os.path.join("uploads", "../pwned_before.txt")` resolves to the parent of `uploads/`, so the file landed at `/app/pwned_before.txt` — outside the intended directory. It worked because untrusted input was used directly to build a filesystem path.
  2. *Why the fix stops it & what could still break it:* `secure_filename()` strips `/`, `\` and `..`, so the name can never climb out of `uploads/`; the allow-list then rejects unexpected types. What could still break it: the process runs as **root** and `uploads/` sits inside `/app`, so a future write-primitive elsewhere is still catastrophic — the class fix needs non-root + storage outside the app dir too.

🤖 Audit the AI

1. AI answer I audited (a common weaker fix an assistant offers for this endpoint):

```
@app.route("/upload", methods=["POST"])
def upload():
    f = request.files["file"]
    if ".." in f.filename:                # block traversal
```

```
        return "bad filename", 400
    f.save(os.path.join(UPLOAD_DIR, f.filename))
    return {"saved": f.filename}
```

2. What's wrong with it (quoted lines):
- `if ".." in f.filename` is a **blocklist**, and it misses the absolute-path bypass. I verified in the container: `os.path.join("uploads", "/etc/passwd")` returns `/etc/passwd` — when the second argument is absolute, `os.path.join` *discards* `"uploads"` entirely. A filename of `/etc/passwd` contains no `..`, passes the check, and still escapes.
- It also misses Windows separators (`\`) and any encoded/normalised variant, and
- `f.save(os.path.join(UPLOAD_DIR, f.filename))` still uses the **raw** filename, and there's **no extension allow-list**, so `shell.php` sails through.

3. Correct, verified version (what I committed, `c254723`):

```
from werkzeug.utils import secure_filename
ALLOWED_EXT = {".txt", ".png", ".jpg", ".jpeg", ".pdf"}
...
safe = secure_filename(f.filename or "")
if not safe or os.path.splitext(safe)[1].lower() not in ALLOWED_EXT:
    return {"error": "rejected filename"}, 400
f.save(os.path.join(UPLOAD_DIR, safe))
return {"saved": safe}
```

The AI's fix was insufficient because it blocklisted one pattern instead of *removing user control of the path component*; `secure_filename` is an allow-list transform (verified: `secure_filename("/etc/passwd") == "etc_passwd"`, `secure_filename("../x.txt") == "x.txt"`), so the whole class closes rather than one spelling of it.

🧠 Comprehension & Prompt

A. Explain in Plain English.
The `/upload` endpoint takes the filename straight from whoever uploads and uses it to build the path where the file is saved. Because it never strips the directory parts, a filename like `../pwned.txt` makes the save path climb out of the uploads folder into the app directory — so an attacker can write or overwrite files anywhere the process can reach. It's exploitable because untrusted input is used directly as a filesystem path.

B. Prompt Problem.
Final prompt that produced a correct, secure fix:

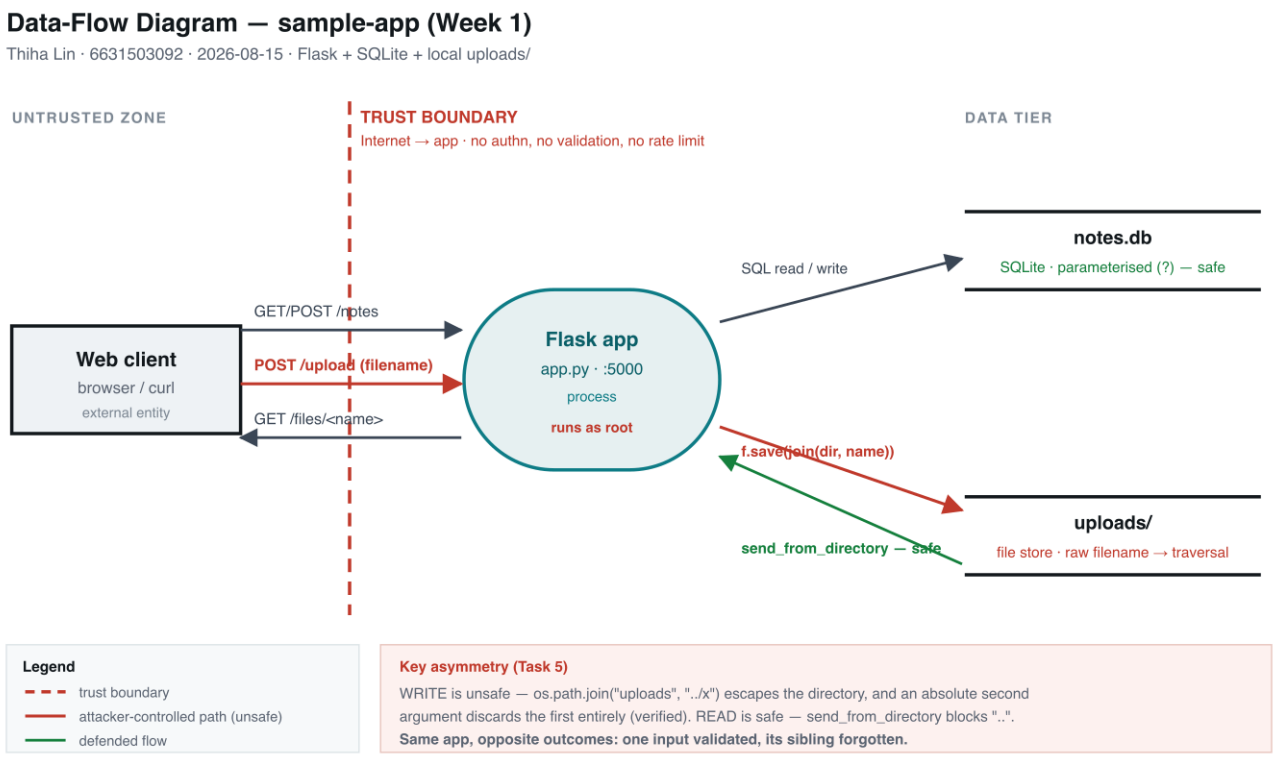
> "This Flask `/upload` handler passes a user-controlled filename to `os.path.join` and saves it, allowing path traversal (`../`) and absolute-path escape. Rewrite the function so **no user-supplied string can become a path component**: use `werkzeug.utils.secure_filename`, reject an empty result, enforce an extension allow-list `{.txt,.png,.jpg,.jpeg,.pdf}`, keep saving under `UPLOAD_DIR`, and return HTTP 400 on rejection. Show only the changed function."

Verified: applying it produced commit `c254723`; re-running `POST /upload filename=../pwned.txt` now returns `{"saved":"pwned.txt"}` and nothing lands in `/app` — exploit fails.

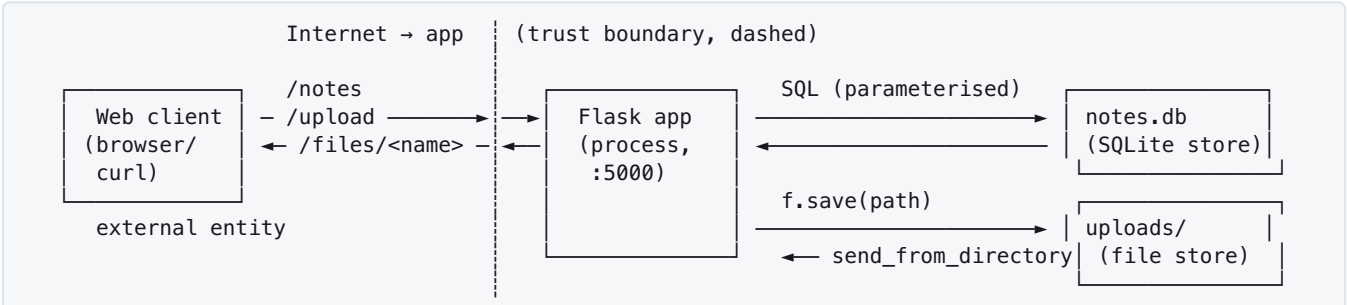
Threat Model — sample-app (Week 1)

Student: Thiha Lin · 6631503092 · 2026-08-15
Target: `labs/week01-threat-modeling/sample-app/app.py` (Flask, SQLite, local `uploads/`)
Runs on: `http://localhost:8081` (host 8080 taken by another local service)
Scope: design analysis; the single before/after in §9 is the worksheet's Task-8 defense check.

1. Data-flow diagram



Structure reference (same content as the image):



Dashed line = the only trust boundary, and it has no check on it. Source: `img/dfd-6631503092.svg`.

2. Elements & trust boundaries

Element	Type	Trust boundary crossed?
Web client	external entity	**yes** — Internet → app. Every request is untrusted input.
Flask app	process	

Element	Type	Trust boundary crossed?
		The boundary enforcement point — and it enforces nothing: no authn, no authz, no validation.
SQLite DB (`notes.db`)	data store	app → data tier. Reached only via the app; queries use `?` placeholders → **no SQLi here.**
`uploads/` store	data store	app → filesystem. **Crossed with attacker-controlled data**: `f.filename` becomes a path component unmodified.

3. STRIDE analysis (full)

Element	S	T	R	I	D	E
/notes	client-supplied `owner`, no auth → post/read as anyone	no ownership check → tamper with any note in the shared table	no logging → no record of who wrote what	`GET /notes` returns **all** users' notes to any caller	unbounded body + ever-growing table, no rate limit → flood to exhaust disk/mem	seed `owner=admin`-style data a higher-trust component later trusts
/upload	no auth on who may upload → anonymous write	**saves raw `f.filename` → arbitrary file write (`../`)**	no logging of uploads	**echoes resolved save path** → leaks filesystem layout	no size/type/rate limit → fill disk	write into `/app` (served/executed dir) → code/content execution
/files/	no auth → anyone reads any served file	read-only endpoint → low tampering risk	no logging of reads	**comparatively defended:** `send_from_directory` → `safe_join` rejects `..` / absolute paths on read (404/403)	repeated large fetches → minor DoS	low alone; serves back whatever `/upload` planted → part of a chain

Facts: no authn/session anywhere · no logging anywhere → Repudiation is universal · no rate/size caps → DoS surface · `notes.db` parameterised → not an injection finding · `/upload` has no `secure_filename()` and no allow-list (before Task 8).

Asymmetry worth noting: the *read* path (`/files`) is safe because `send_from_directory` contains traversal; the *write* path (`/upload`) is not. Same app, opposite outcomes — the classic "one input validated, its sibling forgotten."

4. Task 3b — Systems-level pass

Boundaries end-to-end. One request crosses: (1) **Internet** → **Flask** on entry, and (2) **Flask** → **data store** (`notes.db` *or* `uploads/`) inside. Crossing (1) has **no check at all** — no authentication, no input validation, no rate limit. That single unguarded crossing is the whole security posture of the app.

Flask process fully owned → attacker reaches: every note in `notes.db`, the entire `uploads/` store (read/write/delete), the container filesystem **as root** (Dockerfile has no `USER`), and any network the container can reach. Owning the process = owning all data + the power to serve malicious content back through `/files`.

`uploads/` **fully owned** → because writes escape `uploads/` (the `../` bug), "owning uploads/" is really "owning `/app`": the attacker reaches the app's own source (`app.py`), and anything the process reads at runtime. The store's blast radius is not the store.

Chain (two "low" findings → unacceptable):
`/upload arbitrary write (rated modeling-only, "no exec this week")` → `overwrite /app/app.py (or drop a file the app serves/imports)` → **persistent remote code execution.** Neither step looks fatal in a per-element grid; together they are total compromise.

System claim. *Even if every element-level mitigation in Task 8 is implemented, this system still fails if it keeps running as root with no authentication boundary — because missing authn and root are system properties, not element bugs, so any single future element flaw is again total.*

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5. Task 4 — Abuse cases & attacker personas

Persona A — Mallory (anonymous internet attacker; no account, just reaches the URL).
- **AC-1:** Mallory `POST /upload` with `filename=../app.py` to overwrite the running app code (Tampering → EoP), against the `/upload` flow + Flask process.
- **AC-2:** Mallory `GET /notes` and harvests every user's notes (Information disclosure), against the `/notes` flow + `notes.db`.

Persona B — Trudy (curious low-privilege user; treated as anon here since there's no auth, models insider curiosity).
- **AC-3:** Trudy `POST /notes` with `owner=admin` to plant authoritative-looking data others trust (Spoofing), against `/notes`.
- **AC-4:** Trudy uploads many large files right before a deadline to fill the disk and deny classmates service (DoS), against `/upload` + filesystem.

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6. Task 5 — Path-traversal deep-dive

Data flow. client → `POST /upload` (multipart; `filename` attacker-controlled) → `f.save(os.path.join("uploads", f.filename))`. `os.path.join("uploads","../pwned.txt")` = `"uploads/../pwned.txt"`, which the OS resolves to the **parent** of `uploads/` (= `/app`). Later, `GET /files/<name>` uses `send_from_directory`, which is **safe on read** (`safe_join` blocks `..`).
→ **Write is unsafe, read is safe** — the asymmetry from §3.

Why `../` escapes. `os.path.join` concatenates with a separator and does **not** normalise or contain; a leading `..` walks up, and (verified) an **absolute** second argument discards the first entirely: `os.path.join("uploads","/etc/passwd") == "/etc/passwd"`. Nothing checks the final path stays under `uploads/`.

Secure design. (1) `secure_filename()` to strip separators and `..`; (2) store uploads **outside** the app/web-served dir; (3) allow-list extensions + verify content type/magic bytes; (4) generate a server-side random name, keep the original only as metadata; (5) enforce size limits; (6) run the process **non-root**. **Class invariant:** *no user-supplied string ever becomes a path component.*

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7. Task 6 — NoteVault (term-project target) kickoff

DFD adds, over the sample app: a **login/session** flow (JWT cookie), a `/register` flow, a `/search` flow, an `/admin` flow, and an `/export` flow — every one crossing the same unguarded Internet→app boundary. **Top-3 STRIDE threats to investigate first:**
1. **SQLi login bypass** — `/login` builds the query by string-formatting (Tampering/EoP, CWE-89).

2. **JWT `alg:none` forgery** — decode allow-list includes `"none"`, so an unsigned token is trusted (Spoofing, CWE-347).
3. **Mass-assignment admin** — `/register` accepts a client-supplied `role` (EoP, CWE-269).

(Full 16-finding map lives in the project assessment; these three seed the report.)

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8. Task 7 — Security requirements (acceptance criteria)

1. **The system must** authenticate the caller and bind each note to that identity, **so that** a client cannot create or read notes as another user. *(→ /notes Spoofing + Info disclosure)*
2. **The system must** reject any upload whose sanitized filename differs from the supplied name or whose extension is not in the allow-list, **so that** no user-supplied string becomes a path component. *(→ /upload Tampering)*
3. **The system must** write a timestamped, identity-attributed audit entry for every create / upload / read, **so that** actions are attributable and non-repudiable. *(→ Repudiation, all elements)*

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9. Task 8 — Rank, mitigate, prove

#	Threat	Element	L	I	Score	Mitigation
1	Arbitrary file write via path traversal	/upload	High	High	**Critical**	`secure_filename()` + allow-list — **IMPLEMENTED**
2	Read all users' notes (no authz)	GET /notes	High	High	**Critical**	authn + per-owner query filter
3	Impersonation via client-supplied `owner`	POST /notes	High	Med	High	authn; derive `owner` from session
4	No audit logging (repudiation)	all	Med	Med	Med	structured request logging
5	Disk/CPU exhaustion (no limits)	/upload,/notes	Med	Med	Med	`MAX_CONTENT_LENGTH` + rate limit

Implemented: #1. Commit `c254723` on `wk01`.

Before (unfixed):

```
$ curl -s -X POST localhost:8081/upload -F "file=@demo.txt;filename=../pwned_before.txt"
{"saved":"../pwned_before.txt"}
$ docker exec …-sample-app-1 ls -la /app/pwned_before.txt
-rw-r--r-- 1 root root 13 … /app/pwned_before.txt      ← escaped uploads/, landed in /app
```

After (fix, rebuilt):

```
$ curl -s -X POST localhost:8081/upload -F "file=@demo.txt;filename=../pwned_after.txt"
{"saved":"pwned_after.txt"}                            ← sanitized, stays in uploads/
$ docker exec …-sample-app-1 ls /app/pwned_after.txt
ls: cannot access '/app/pwned_after.txt': No such file or directory   ← escape blocked
$ curl … filename=notes.txt   → {"saved":"notes.txt"}   ← legit upload still works
$ curl … filename=shell.php   → {"error":"rejected filename"}  ← allow-list rejects
```

Class vs instance. This is an **instance** fix (it hardens `/upload`). The **class** fix is the invariant "*no user-supplied string ever becomes a path component*" — applied everywhere, plus

running non-root and storing uploads outside `/app` so that even a future write-primitive can't reach code. `secure_filename()` closes this endpoint; the invariant closes the class.